

Example 19: Can we tile a board obtained by removing one of the four corner squares of a standard checkerboard?

Solution: If we remove one corner square, we will have 63 squares. If we tile the checkerboard now, we can at most tile 62 squares with 31 dominoes. One square will be left over, since a domino has 2 squares. $\therefore$ The board cannot be tiled $\square$

Example 20: Can we tile the board obtained by deleting the upper left and lower right corner squares of a standard checkerboard, shown in Figure

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B	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W

Figure 5: The standard checkerboard with the upper left and lower right squares removed.

Solution: From Example 18, we have seen that 32 dominoes can tile a 8×8 checkerboard. Since each domino has 1 white square and 1 black square, there are a total of 32 white squares and 32 black squares. On removing the 2 corner squares as shown, we have $32 - 2 = 30$ white squares and 32 black squares remaining (Wlog, assume the upper left and lower right corner squares are white). Now, if we tile intelligently, we can use 30 dominoes to cover 30 white squares and 30 black squares.